

TEST PROCESS

We broadly categorised the cards into two groups: high-performance and general use. We tested the graphics cards on a high-end machine and then on a low-end machine in order to get a clear idea of how a high-end graphics card would perform on a low-end system as well as on a very powerful machine. Will it deliver a real performance boost by taking the load off the processor? We also took price and features into account and rated the cards accordingly.

To test the performance of the cards,

we ran the 3DMark 2000 and the 3DMark 2001 tests, and the Vulpine GL Mark test, which is a very graphics-intensive test and specifically tests the OpenGL capabilities of the card. We also ran a timedemo in *Quake III* in normal, high and max settings, and *Serious Sam*, a game which is highly configurable for graphics and visual effects. Another test run on the graphics cards was *Evolva*. This benchmark runs a demo game with bump mapped textures and displays the average frame rate.

speed gameplay. And this is what the newest range of 3D cards promise to offer.

Only a high-end gamer would appreciate these qualities. Those who are into 2D or low-end 3D gaming would do just fine without these high-end cards. And who would need Full Screen Anti Aliasing to run office suites anyway?

Such is the power of today's graphics processors, that a new name has been coined for them—the Graphics Processing Unit (GPU). Looking at where GPU technology is heading, who knows, in the future the drops of blood flying around from your freshly fragged bot might have trillions of polygons each, thus giving you a feel of playing the game with movie-quality special effects.

Name your game

To judge which kind of user a particular card would be suitable for, we categorised gamers according to the kind of games they play and the kind of features/gameplay they would be happy with. Accordingly, a hardcore gamer would be someone who likes to play games at high resolutions (at least 1024x768x32) with the extra effects turned on. The machine used by such a gamer would also be a high-end system—a P-III 1 GHz with at least 128 MB of RAM.

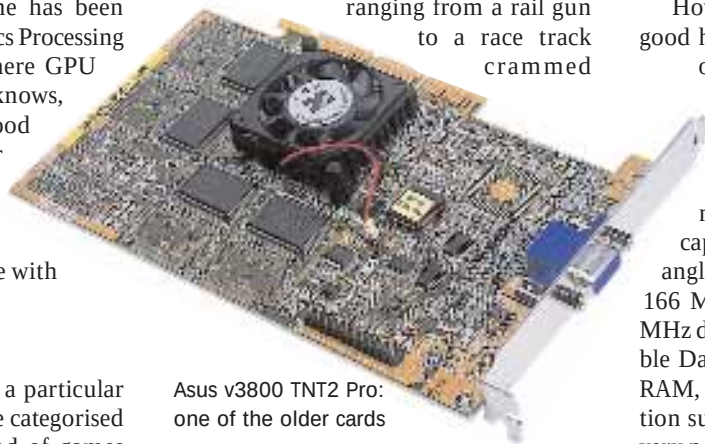
An average gamer would be someone who plays games at a fairly high resolution (800x600x32) but with some of the extra frills off and a casual gamer would be someone who is okay with the idea of playing games at the least possible resolution (640x480x16) and with all the effects turned off.

High-end test

Hardcore is the word

Are you the kind who think that blood should look like blood as your opponent takes a hit when you rail him and get as much satisfaction from the realistic effects as from the gameplay itself? Then you should settle for nothing less than the

Asus V8200 GeForce3 Deluxe with 64 DDR memory to swap textures ranging from a rail gun to a race track crammed



Asus v3800 TNT2 Pro: one of the older cards

with sports cars. The card features the GeForce3 chipset, which as a GPU is loaded with almost every technologically advanced feature to keep the hardcore gamer happy until the next advanced chipset is released. What's more, the card, with its nFiniteFx Engine, leaves space for game programmers to include new features in the upcoming games.

The scores logged by this card in each of our tests will simply amaze you. The 3D Mark 2000 and 2001 scores prove the fact that this card is simply the best as far as Direct 3D performance is concerned. In 3DMark 2001 it notched up a never-before-recorded score of 5751. The card excelled in the OpenGL benchmarks

as well. In the Vulpine GL Mark test we recorded a score of 47.7 fps with max settings. As far as performance is concerned, this card is a true winner and the best choice to compliment your high-end machine.

Not very close behind is the Asus V7700 GeForce2 Ultra. Hardcore gamers would appreciate the card's capability of pumping out 31 million triangles per second and its huge memory bandwidth of 7.36 GBps—this means that you can crank up all those texture settings in your games without having to worry about your gameplay slowing down. The card's memory clock runs at a supercharged 460 MHz DDR, and to keep the heat under control you will find two strips of heat sink on the RAM chips on each side of the card.

On our high-end machine, the card was simply outstanding. In *Quake III* it outperformed the GeForce3 in the normal mode by giving an average frame rate of 220.5 fps. One of the reasons for this is the higher core frequency of the card (250 MHz). If you are stuck for a good card with all the GeForce3-based cards sold out, then this card is what you should buy.

However, if you want to buy a really good high-end card and are falling short of cash, then you could go in for the Asus V7700 GeForce2 Pro Deluxe, featuring the commendable GeForce2 GTS chipset but clocked at a higher core, which means the same core but with the capability to churn out 25 million triangles per second. Its memory speed is 166 MHz but it effectively runs at 333 MHz due to the fact that it has DDR (Double Data Rate) memory. With 64 MB of RAM, playing *Quake III* at a high resolution such as 1280x1024 at 32-bit depth is very possible without worrying about the smooth gameplay and the eye candy. The lower price (Rs 29,000 vs Rs 38,000 for the Asus V7700 GeForce2 Ultra) is the icing on the cake.

The AOpen GeForce2 GTS DDR is also worth a mention here. This card too is based on a GeForce2 GTS graphics processor, which ensures very good high-end gaming because of the fast processor and also because it features 32 MB of DDR SGRAM. The scores of this card were expectedly impressive, especially considering its low price. If you want a value solution in the high-end category then this card is your best bet.

The Elsa Synergy3 Quadro2 MXr performed well in 3DMark 2000 with a